

THEEGELA VAMSI KRISHNA

Electronics Engineering (VLSI) Undergraduate | ASIC Design | Semiconductor Research

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Professional Summary

Third-year Electronics Engineering (VLSI) undergraduate at Manipal Institute of Technology with hands-on experience in RTL and ASIC design, computer architecture, custom CMOS, SRAM characterization, semiconductor TCAD, and physical design. Experienced with Cadence and Synopsys EDA tools and open-source RTL-to-GDSII flows. Currently pursuing B.Tech (Honours) in Photonics.

Research & Experience

VLSI IC Design Intern, *ASD Academy of Skill Development* 2026 – Present

- Designed a 2×2 MAC-based systolic-array accelerator in Verilog.
- Developed RTL designs and digital systems using Verilog and VHDL.
- Gained exposure to CMOS layout, RTL design, FSMs, timing analysis and ASIC methodology.

Undergraduate Research Assistant, *MIT Manipal* Ongoing

Supervisor: Dr. Arjun Sunil Rao

- Designing and comparatively analyzing 6T and 8T SRAM cells under PVT and Monte Carlo variations using Cadence Virtuoso.
- Performing schematic and layout design, DRC/LVS, parasitic extraction, SNM, delay, leakage and power characterization.
- Conducting research on logic locking and programmable delay logic for hardware-security applications.

Industrial Training, *Ardent Software* 06/2026 – 08/2026

Completed industrial training in VLSI-Based IC Design.

- Developed a Digital Door Lock System using VHDL.

Education

B.Tech in Electronics Engineering (VLSI) | B.Tech (Honours) in Photonics, 2028

Manipal Institute of Technology

Manipal

CGPA: 8.22/10

Technical Skills

HDLs

Verilog, SystemVerilog, VHDL

Cadence

Virtuoso, Spectre, Xcelium, Genus, Innovus, Voltus, Quantus

Open-Source VLSI

Yosys, OpenLane, OpenROAD, OpenSTA, Magic, KLayout

Programming

C, Embedded C, Python, MATLAB, TCL, Bash

TCAD & Simulation

Synopsys Sentaurus TCAD, COMSOL Multiphysics, Aestimo 1D, LTspice, ngspice

Hardware

KiCad, FPGA Development-Artix7 Basys3, Embedded Systems, CAN Electronics, Sensor Integration

Research and Projects

Photonic Bandgap Engineering Using Distributed Bragg Reflectors

Ongoing

Developing multilayer photonic structures in COMSOL Multiphysics to investigate wavelength-selective reflectance, refractive-index contrast and photonic bandgap behaviour.

RISC-V SoC with DMA & Error-Tolerant Computing

Designed an RTL-level RISC-V computing architecture integrating DMA functionality and error-tolerant arithmetic hardware, demonstrating experience in computer architecture, RTL design and system-level hardware integration.

Approximate Multiplier for Energy-Efficient Computing

Designed and evaluated truncated and compensated multiplier architectures for PPA-accuracy trade-offs. Achieved approximately 5.1% area reduction, while the compensated architecture achieved approximately 50% error reduction compared with the truncated multiplier with comparable timing performance.

TCAD-Based HVK SJ-LDMOS Device Analysis

Performed semiconductor process and device simulation using Synopsys Sentaurus SProcess, SDevice, SVisual and Inspect. Analyzed electrical characteristics, doping profiles, carrier distribution, electric fields and device performance.

Patent & Achievements

Hardware Architecture for Real-Time Tire Grip Estimation and Low-Latency

2026

Motor Torque Control for Lightweight Electric Vehicles

Provisional Indian Patent Application No. 202641053073

IEEE Student Member

IEEE GRSS Member

NCC Corporal

Alpha Grade | NCC 'B' Certificate

Team Karting Manipal

Electronics & Embedded Systems Team

MIT Manipal Hockey Team

Inter-MAHE Winners: 2024, 2025 & 2026